

# The Importance of Explainable AI in the Era of Agentic Systems

Explainable AI (XAI) makes AI systems transparent, explainable, and credible. It ensures users understand why a given model arrived at a specific decision so that it is easier to verify, debug, and trust the system. Let's look at all that XAI entails followed by a real-world example of how it works.



**A**gentic AI represents the next evolution of intelligent systems—digital entities that not only follow instructions but also set goals, make decisions, and learn independently. These agents will soon encompass every aspect of human life, from finance and energy to robotics and personalised courses. But with this exciting potential comes a critical challenge: understanding what these agents are thinking. Agentic AI does not simply follow conventional systems but engages in self-directed behaviour influenced by constantly changing goals, interaction with the environment, and acquired knowledge, thereby giving rise to some very complicated and often opaque processes behind decisions. An example: A totally autonomous vehicle may suddenly make an unexpected manoeuvre. Or a healthcare agent may recommend some treatment that is not of common standard protocol. Certainly, knowing what an AI agent has done is insufficient. We want to know why it did so.

But why is this understanding so vital?

First, we are more likely to trust technologies we understand. If agentic systems are black boxes making

decisions without clear rationale, it will lead to erosion of public trust and subsequently hinder the spread of such systems, and the benefits derived from them. Would you trust a rationally opaque entity to control your health or finances?

Second, it is something that must be done. Since agents are going to take on new roles, accountability is primary. If an autonomous agent inflicts damage or makes a blunder, who comes under fire when there is no knowledge of the system's mental processes?

It's also important to know why an agent made a certain decision so that its logic may be debugged and risks mitigated to ensure it operates safely and reliably.

Finally, we need this understanding for further growth and improvement. Knowledge of the strengths and weaknesses of an agent's reasoning enables algorithm refinement and helps build better systems. We learn from their successes and failures.

As intelligent agents take on greater roles in society, understanding their decision-making is no longer optional—it's essential for trust, accountability, and safety. To fully harness their potential, we must ensure their actions are transparent and explainable. This is the critical role of explainable AI (XAI), which enables us to interpret and trust these increasingly autonomous systems.

By giving results without informing how it gets them, AI, especially deep learning, often works as a 'black box'. Explainable AI is designed to fulfil the goal of rendering AI decisions evident and understandable to humans -- it helps understand why an AI system came to a particular conclusion.

The methodologies explainable AI uses are briefly outlined below.